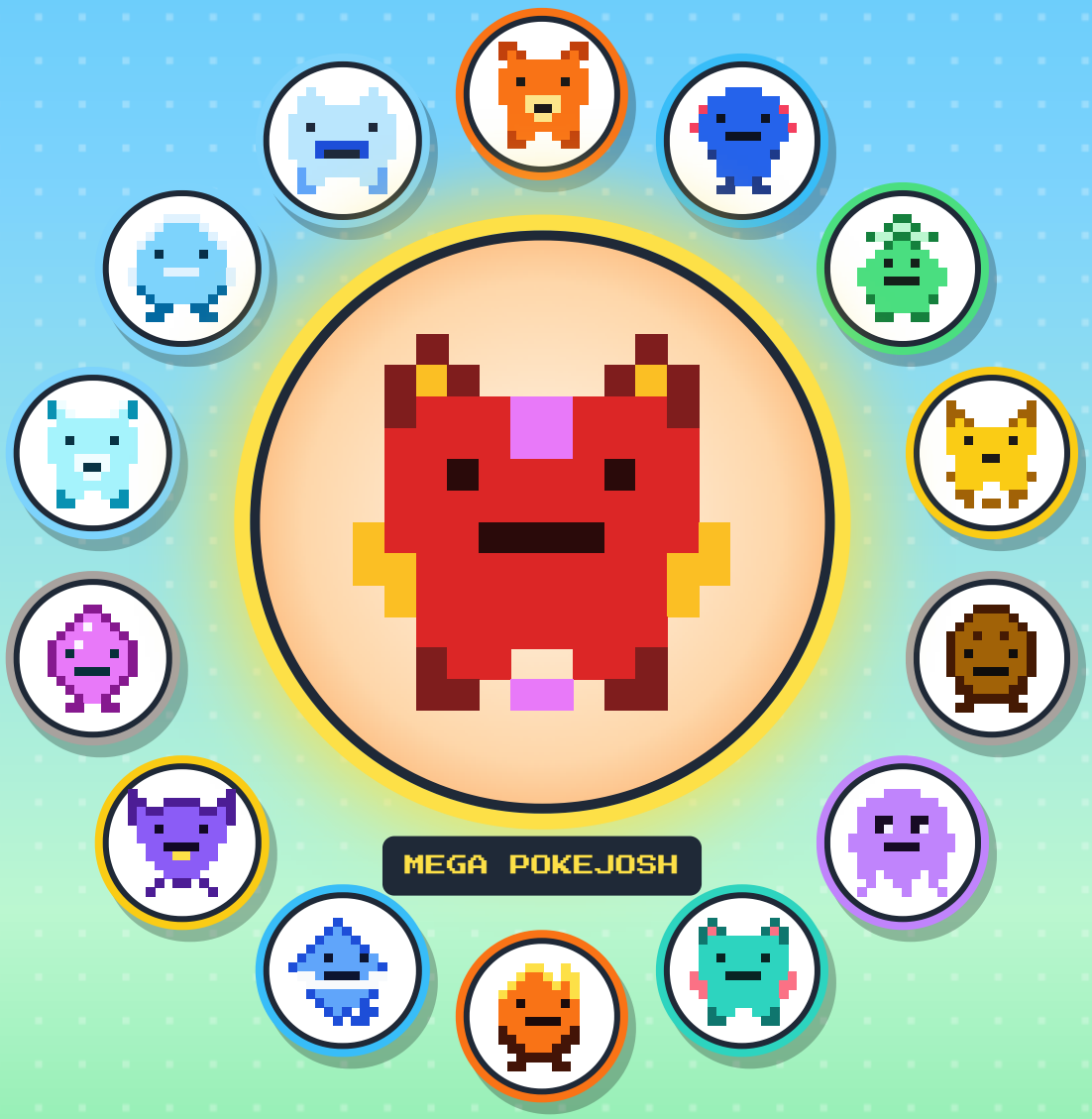
OFFICIAL

MATHIMAL QUEST

Field Guide & Trainer's Manual



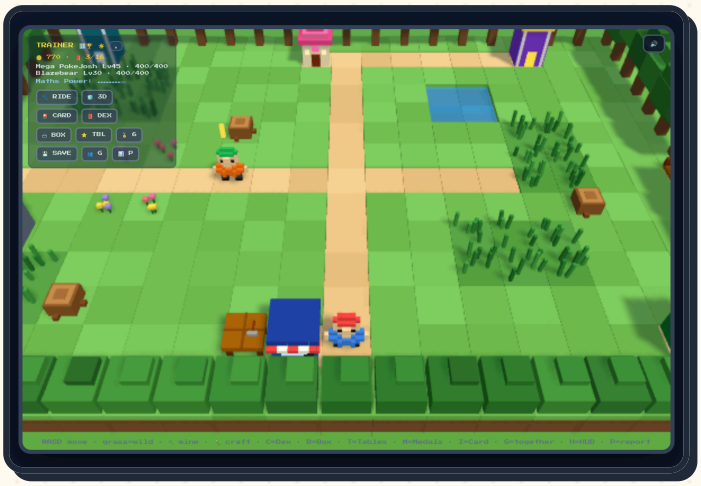
This guide belongs to Trainer: _____

Every Mathimal · every gym · catching · crafting · medals · secrets of the Clock Tower

For game version v5.0

Welcome, Trainer!

You are a young trainer in a world where creatures called **Mathimals** grow stronger with every maths question you answer. Explore three regions, catch them all, win badges — and become **Grand Champion**.



The 3D world. Your HUD (top-left) shows coins, team and Maths Power.

🗺️ Your quest

1. **Pick a partner** — Embercub, Bublip or Thornling.
2. **Beat Professor Prime** for the **Number Badge**. You'll get **PokeJosh** as a gift!
3. **Cross into the Ember Caves** and beat **Captain Magna**.
4. **Catch 8 species** to open the **Frozen Peaks**.
5. **Face the Elite Four** and **Grandmaster Aria**.
6. **Win all 4 Topic Badges**, climb the Tower and collect every medal!

Keyboard controls

🌳 In the world

W A S D or arrow keys	Walk around
C	Mathi-Dex (who you've seen & caught)
B	Box — swap Mathimals in and out of your team
T	Times-table stars
M	Medals & Topic Badges
I	Trainer Card
P	Training report
H · G	Hide the HUD · Play together
Esc	Close any window

⚔️ In a battle

F then F	Fight → Quick attack (or 1 Quick, 2 Power)
C then C	Catch → throw a Math Ball (1–7 pick a ball)
B	Bag (potions)
S	Switch Mathimal (1–4)
R	Run (Y yes / N no)
type + Enter	Answer a number question
1–4	Pick a multiple-choice answer
P	Pause (nothing happens while paused)
Esc	Go back a menu

💡 **Tip:** Every button in a battle shows its key in the corner — you never need the mouse!

How Battles Work



Choose what to do — every action needs a maths answer.

🔁 A turn, step by step

1. Choose **FIGHT**, then a move.
2. A maths question appears — answer it before the timer runs out.
3. **Right?** Your Mathimal attacks! **Wrong?** The attack misses.
4. Then the other Mathimal hits back.

★
Quick move
easier question, normal damage

★★
Power move
harder question, **×1.6** damage

⚡ Speed bonus

Answer **fast** (under 9 seconds, or 12 for multiple choice) and your attack does **×1.3** damage.

🔥 Combo meter

Right answers in a row build a combo — up to **×1.5** damage. A wrong answer resets it.

📈 Maths Power

4 quick, correct answers in a row raise your Maths Power (1–10). Higher power = harder questions = stronger trainer. 2 misses in a row lower it.

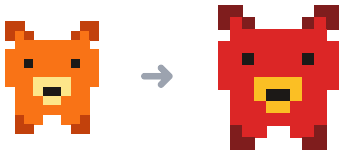
Type chart

ATTACK ↓ / DEFEND →	🔥	💧	🌿	⚡	🪨	👻	🐉	❄️	🔮
🔥 Fire		×¾	×1.5		×¾			×1.5	
💧 Water	×1.5		×¾	×¾	×1.5				
🌿 Grass	×¾	×1.5			×1.5			×¾	
⚡ Electric		×1.5			×¾				
🪨 Rock	×1.5	×¾	×¾	×1.5				×1.5	×¾
👻 Ghost									×¾
🐉 Dragon								×¾	
❄️ Ice	×¾		×1.5		×¾		×1.5		
🔮 Psychic					×1.5	×1.5			

Green ×1.5 = super effective. Red ×¾ = not very effective. Pick the right Mathimal for the job!

✨ Evolution

At **Level 8** every Mathimal evolves into a bigger, stronger form with new colours.



Embercub → Blazebear

💗 Fainting & healing

If your whole team faints, you rush home and everyone is healed. Step on a **Heal House** 💗 any time to heal for free, or use a Potion from your Bag.

🎯 Catching Mathimals

Walk through **tall grass** 🌿 to meet wild Mathimals. To catch one, choose **CATCH**, pick a ball and answer the question correctly.



Correct answer — the ball flies!

📏 The golden rules

- ✔ You can only catch a Mathimal once its HP is **below 70%** — weaken it first.
- ✔ The **weaker** it is, the easier it is to catch.
- ✔ A wrong answer means the ball misses.
- ✔ You can't catch a trainer's Mathimals.
- ✔ **Shiny** ✨ Mathimals (1 in 20) have special colours. Collect them!

Catch chances (answer correctly, then...)

Wild Mathimal at **70%** HP
- Math Ball 35%
- Great Ball 60%

Wild Mathimal at **50%** HP
- Math Ball 50%
- Great Ball 75%

Wild Mathimal at **30%** HP
- Math Ball 64%
- Great Ball 89%

Wild Mathimal at **10%** HP
- Math Ball 80%
- Great Ball 98%

Legendary Mathimals are twice as hard to catch. The Mega Ball always works.

The ball family

Math Ball
Always free
The basic ball. Unlimited!

Great Ball
+25%
Shop 25🪙 or craft

Dusk Ball
+40% at night
Almost useless by day. Craft

Net Ball
+40% vs 💧 & ❄️
For water & ice types. Craft

Timer Ball
+6% per turn
Up to +40% in long battles. Craft

Heal Ball
+15%
Caught Mathimal is fully healed. Craft

MEGA BALL
NEVER fails!
Save it for something special. Craft

🎒 Team & Box

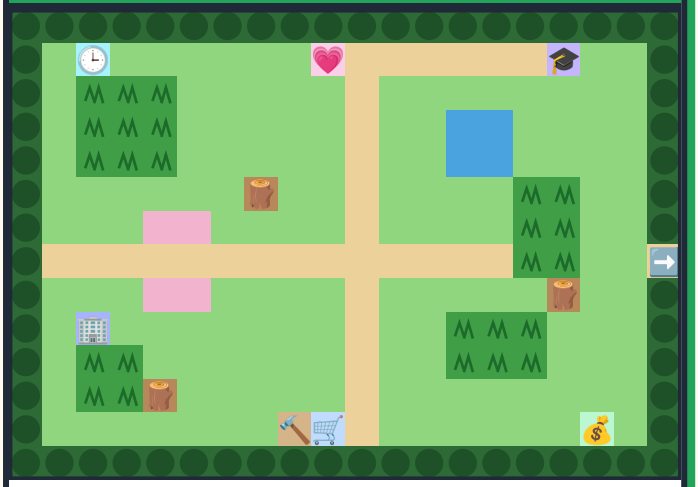
You carry up to **4** Mathimals. New catches go to your **Box** (holds 400). Press **B** to swap.

🐾 Riding

A healthy Mathimal at **Lv20+** can carry you! Riding is faster, and wild Mathimals leave you alone. Press **RIDE** on the HUD.

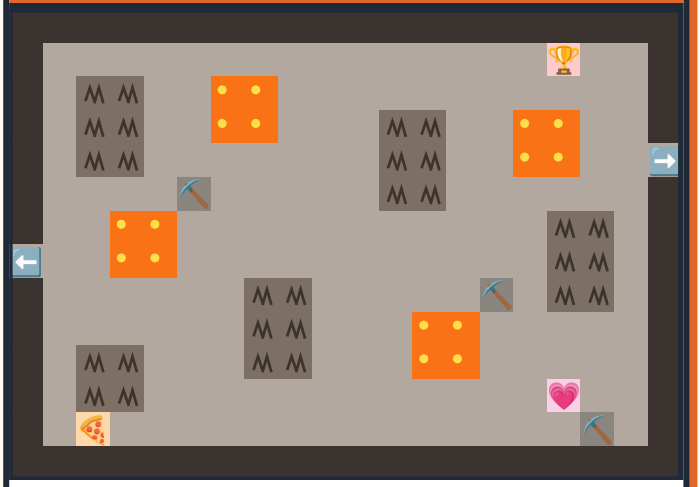
The World

The Meadow — Start here



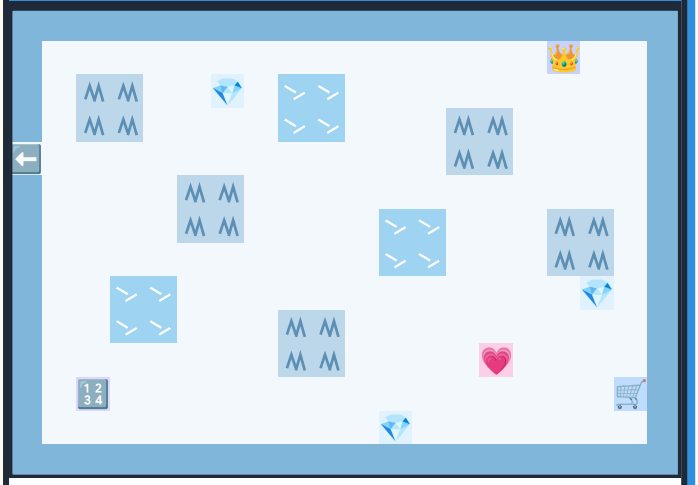
Grassy fields. Gym 🎓, Clock Tower 🕒, Coin Gym 💰, Trainer Tower 🏢, shop and crafting bench.

Ember Caves — 🔒 Number Badge



Hot rocks and lava (don't walk on it!). Gym 🏆, Fraction Forge 🍕, ore to mine.

Frozen Peaks — 🔒 Both badges + 8 species



Snow and frozen lakes. The Elite Four 👑, Number Vault 🔢 and rare ice Mathimals.

🔑 Map key

🎓	Professor Prime's gym
🏆	Captain Magna's gym
👑	Elite Four hall
💰🕒🍕🔢	Topic gyms
🏢	Trainer Tower
💗	Heal House
🛒	Mathi-Mart shop
🔨	Crafting bench
🪵⛏️💎	Mine wood, ore & crystal
➡️	Bridge to next region
🟩🟫🟦	Tall grass — wild Mathimals!

🧭 Explorer tips

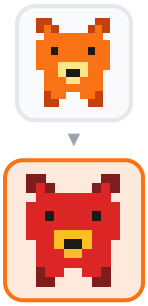
- ✔ Walk **into** a building to go inside.
- ✔ Water, lava and frozen lakes are walls — find a way around.
- ✔ Wild Mathimals only hide in **tall grass**.
- ✔ Rival Max ⚡ waits on the path — you'll know when you meet him!

🌙 Day & night

Every 80 steps, day turns to night (and back). Different Mathimals come out at night — and Dusk Balls get super strong.

The Mathi-Dex · 1

All 16 Mathimals. The **%** is the chance that a wild meeting in that place and time is this Mathimal. Tick the boxes as you go!



Embercub → Blazebear

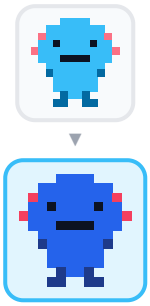
#001

FIRE · ❤️ 24 HP · 💥 6 ATK

★ Ember Dash · ★★ Flame Burst

Meadow	☀️ day	Uncommon	11%
Ember Caves	☀️ day	Uncommon	15%

☐ seen ☐ caught ☐ shiny



Bublip → Tidemaw

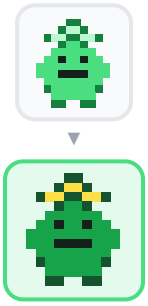
#002

WATER · ❤️ 27 HP · 💥 5 ATK

★ Bubble Pop · ★★ Tidal Splash

Meadow	☀️ day	Common	24%
Frozen Peaks	☀️ day	RARE	6%

☐ seen ☐ caught ☐ shiny



Thornling → Bramblion

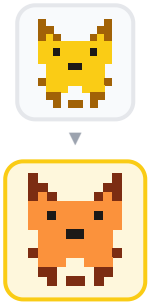
#003

GRASS · ❤️ 26 HP · 💥 5 ATK

★ Leaf Flick · ★★ Vine Whirl

Meadow	☀️ day	Common	24%
Meadow	🌙 night	Uncommon	15%

☐ seen ☐ caught ☐ shiny



Sparkit → Voltrix

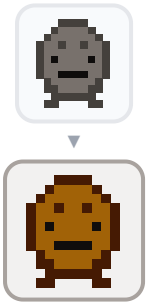
#004

ELECTRIC · ❤️ 22 HP · 💥 6 ATK

★ Static Zap · ★★ Thunder Pounce

Meadow	☀️ day	Common	24%
Meadow	🌙 night	Common	32%

☐ seen ☐ caught ☐ shiny



Rockle → Boulderon

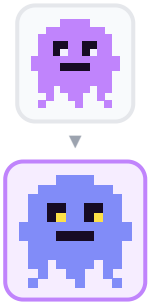
#005

ROCK · ❤️ 32 HP · 💥 4 ATK

★ Pebble Toss · ★★ Boulder Roll

Meadow	☀️ day	Uncommon	11%
Meadow	🌙 night	Uncommon	15%
Ember Caves	☀️ day	Common	32%

☐ seen ☐ caught ☐ shiny



Wispur → Phantomane

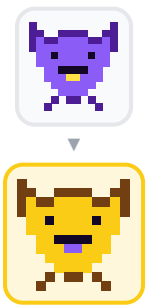
#006

GHOST · ❤️ 21 HP · 💥 7 ATK

★ Spook Tap · ★★ Shadow Swirl

Meadow	☀️ day	RARE	5%
Meadow	🌙 night	Common	32%
Ember Caves	🌙 night	Uncommon	15%

☐ seen ☐ caught ☐ shiny



Batvolt → Thunderwing

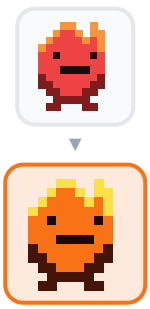
#007

ELECTRIC · ❤️ 23 HP · 💥 7 ATK

★ Echo Zap · ★★ Storm Wing

Meadow	🌙 night	RARE	6%
Ember Caves	🌙 night	Common	32%
Frozen Peaks	🌙 night	Uncommon	15%

☐ seen ☐ caught ☐ shiny



Magmite → Magmagore

#008

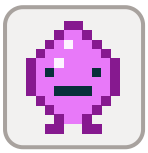
FIRE · ❤️ 28 HP · 💥 6 ATK

★ Spark Spit · ★★ Lava Lob

Ember Caves	☀️ day	Common	32%
Ember Caves	🌙 night	Common	32%

☐ seen ☐ caught ☐ shiny

The Mathi-Dex · 2

Crystix → Crystalon

#09

ROCK ❤️ 30 HP · 💥 5 ATK

★ Shard Shot · ★★ Crystal Crush

Ember Caves	☀️ day	Uncommon	15%
Ember Caves	🌙 night	Uncommon	15%
Frozen Peaks	☀️ day	Uncommon	15%

☐ seen ☐ caught ☐ shiny




Finblade → Finrazor

#10

WATER ❤️ 25 HP · 💥 7 ATK

★ Fin Slash · ★★ Riptide Rush

Ember Caves	☀️ day	RARE	6%
Ember Caves	🌙 night	RARE	6%

☐ seen ☐ caught ☐ shiny




Frostpup → Frostfang

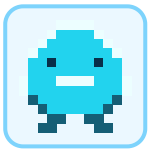
#11

ICE ❤️ 25 HP · 💥 6 ATK

★ Frost Nip · ★★ Ice Fang

Frozen Peaks	☀️ day	Common	32%
Frozen Peaks	🌙 night	Common	32%

☐ seen ☐ caught ☐ shiny

Glacior → Glaciatron

#12

ICE ❤️ 33 HP · 💥 5 ATK

★ Snow Puff · ★★ Blizzard Blast

Frozen Peaks	☀️ day	Common	32%
Frozen Peaks	🌙 night	Common	32%

☐ seen ☐ caught ☐ shiny




Yetai → Yetaint

#13

ICE ❤️ 30 HP · 💥 7 ATK

★ Snowball · ★★ Avalanche

Frozen Peaks	☀️ day	Uncommon	15%
Frozen Peaks	🌙 night	Uncommon	15%

☐ seen ☐ caught ☐ shiny




Drakelet → Dragonyx

#14

DRAGON ❤️ 34 HP · 💥 7 ATK

★ Tail Snap · ★★ Dragon Gust

Frozen Peaks	🌙 night	RARE	6%

☐ seen ☐ caught ☐ shiny




PokeJosh → Mega PokeJosh

#15

PSYCHIC ❤️ 30 HP · 💥 8 ATK

★ Mind Flare · ★★ Psy Inferno

🎁 A gift from Professor Prime after you win the **Number Badge**. Designed by a real young trainer!

☐ seen ☐ caught ☐ shiny




Aurion → Aurion

#16

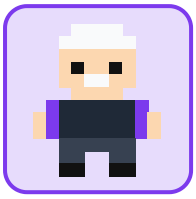
DRAGON ❤️ 42 HP · 💥 9 ATK

★ Golden Claw · ★★ Solar Roar

⚡ Appears in any tall grass (8% of meetings) once you've caught **all 14** regular species. Twice as hard to catch!

☐ seen ☐ caught ☐ shiny

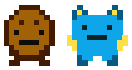
Gyms, Rivals & Champions



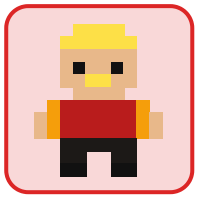
Professor Prime

📍 Meadow — top right 🎓

The first gym leader — and the professor who gave you your partner. His Rockle and Drakelet hit hard, so train up first!



🏅 **Number Badge 🔢 + PokeJosh!**



Captain Magna

📍 Ember Caves — top right 🏆

A fiery leader with a Magmite, a Crystix and a dragon. Bring water and rock types!



🏅 **Champion badge 🏆**

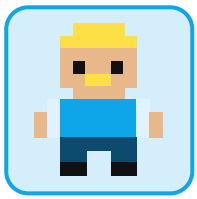


Rival Max

📍 Pops up on the path ⚡

Professor Prime's nephew. He thinks he's the best — he challenges you three times as the story goes on.

🏅 **Coins + bragging rights**



The Elite Four & Grandmaster Aria

📍 Frozen Peaks — the hall 👑

Beat Elite Frost, Elite Shade and Elite Blaze in a row, then Grandmaster Aria. No rest between battles!



🏅 **GRAND CHAMPION 👑**

🏢 Trainer Tower

Meadow, left side. An **endless** climb — each floor is tougher. Win and choose: climb higher, or **leave & heal** and keep your coins. Reward: 10 coins × floor number. How high can you go?

My best floor: _____

⚡ The legend of Aurion



A golden dragon that only wakes when a trainer has caught **every one of the 14** regular Mathimals. Then it can appear in any tall grass... Have your best balls ready!

👑 The Elite gauntlet — four battles in a row

1. **Elite Frost** — Ice



→ 2. **Elite Shade** — Ghost & Electric



→ 3. **Elite Blaze** — Fire



→ 4. **Grandmaster Aria** — The final boss!



Your team's HP carries over between battles — bring Potions and a strong Fire or Rock Mathimal for the ice users!

Rematches: once you've won, leaders can be fought again — their teams are stronger the second time.

The Four Topic Gyms

Four special gyms test a **new kind of maths**. They open after the Number Badge. Inside, every question is about that topic. Win the badge and those questions start appearing in wild battles too!

Coin Gym

📍 Meadow — bottom right, by the shop



Banker Bea

LV 1	$1 and $2 coins · change from $10
LV 2	10c, 20c, 50c coins · change from $20 · "3 treats at $5"
LV 3	Dollars and cents: $3.60 + $1.80 · change from $10

Win the **Coin Badge**

Clock Tower

📍 Meadow — top left corner



Tick-Tock Theo

LV 1	o'clock and half past
LV 2	quarter past and quarter to · minutes in an hour
LV 3	every 5 minutes (4:35) · "what time in 20 minutes?"

Win the **Clock Badge**

Fraction Forge

📍 Ember Caves — bottom left



Chef Fran

LV 1	half of a number · what fraction is coloured?
LV 2	quarters and thirds · ¼ of a pizza
LV 3	¾ of 12, ⅔ of 9 · which fraction is bigger?

Win the **Slice Badge**

Number Vault

📍 Frozen Peaks — bottom left



Professor Digit

LV 1	tens and ones · 10 more / 10 less · count by 2s, 5s, 10s
LV 2	hundreds · round to the nearest 10 · patterns
LV 3	thousands · round to 100 · doubling patterns

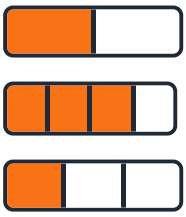
Win the **Digit Badge**

Reading a clock



Short red hand = hour. **Long blue hand** = minutes. Each number is 5 minutes.

Fractions



1/2 — one of 2 parts

3/4 — three of 4 parts

1/3 — one of 3 parts

¼ of 12 → share 12 into 4 groups → **3**. ¾ of 12 → three groups → **9**.

NZ money



100 cents = $1.
Change = what you paid − what it cost.
$10 − $6.40 = **$3.60**

⬆️ **Topic levels:** every topic starts at Level 1. After **8** right answers in that topic it moves to Level 2, and after **20** to Level 3. Check your levels on the Medals screen (M).

Shop, Mining & Crafting

Mathi-Mart (shop)

Potion	+25 HP	15
Super Potion	full HP	40
Great Ball	+25% catch	25
Mystery Egg	hatches after 40 steps · 10% shiny · carry 3	250
Lucky Egg	double XP for 10 wins	300
XP Share	team-mates get half XP (one-time)	1500

Earn coins by winning battles, catching Mathimals and earning medals.

Mining

Wood	Ore	Crystal
tree stumps	rock piles	ice crystals
Meadow	Ember Caves	Frozen Peaks

Walk into it and answer a maths question to mine. It regrows after 60 steps.

Outfits

Classic	Forest Ranger	Midnight	Lava Hunter	Golden Champion
free	400	600	800	5000

Crafting bench recipes

Potion ×3	Great Ball ×2 ×2	Dusk Ball ×2 ×1	Net Ball ×3 ×1
Timer Ball ×1 ×3	Heal Ball ×2 ×2	Super Potion ×2 ×2	MEGA BALL ×2 ×4 ×4

The crafting bench is next to the shop in the Meadow.

Secret codes: grown-ups have special reward codes you can type in at the shop counter. Ask nicely!

Medal Collection

There are **35 medals**. Each one pays coins the first time you earn it. Colour in each medal when you get it!



First Answer
Answer a question correctly
10



Century
100 correct answers
50



Maths Machine
500 correct answers
150



Maths Legend
1,000 correct answers
300



On Fire
10 correct in a row
40



Unstoppable
25 correct in a row
100



Lightning Brain
25 correct answers in under 5 seconds
60



Star Pupil
Master a times table
30



Table Titan
Master 6 times tables
100



Times-Table Champion
Master all 11 times tables
250



Rising Star
Reach Maths Power 5
50



Maths Power MAX
Reach Maths Power 10
200



Money Master
Win the Coin Badge
40



Time Keeper
Win the Clock Badge
40



Fraction Hero
Win the Slice Badge
40



Number Wizard
Win the Digit Badge
40



Topic Grand Slam
Win all 4 Topic Badges
150



Topic Pro
Reach the top level in any topic (20 correct)
75



Number Badge
Beat Professor Prime
30



Champion
Beat Captain Magna
50



Grand Champion
Beat Grandmaster Aria
200



Rival Crusher
Beat Rival Max twice
50



Tower Climber
Clear 5 Tower floors
50



Sky Scraper
Clear 15 Tower floors
150



Battle Veteran
Win 100 battles
100



Gotcha!
Catch a wild Mathimal
10



Super Catcher
Catch 50 wild Mathimals
100



Collector
Catch 8 different species
50



Dex Complete
Catch every regular species
200



Golden Legend
Catch Aurion
200



Shiny Hunter
Own a shiny Mathimal
75



Strong Bond
Raise a Mathimal to Lv30
60



Mega Trainer
Raise a Mathimal to Lv50
150



Miner
Mine 20 times
30



Crafter
Craft 10 items
40

My Trainer Tracker

⭐ Times-table stars

Get **10** right in a table to master it. Colour a box for each one (press **T** in the game to check).

×2	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐ ☆
×3	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐ ☆
×4	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐ ☆
×5	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐ ☆
×6	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐ ☆
×7	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐ ☆
×8	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐ ☆
×9	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐ ☆
×10	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐ ☆
×11	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐ ☆
×12	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐ ☆

🎖️ Badges

- 🔢 Number Badge ☐
- 🏆 Champion ☐
- 👑 Grand Champion ☐
- 💰 Coin Badge ☐
- 🕑 Clock Badge ☐
- 🍕 Slice Badge ☐
- 🔢 Digit Badge ☐

📊 My records

Highest Maths Power: _____

Best Tower floor: _____

Longest answer streak: _____

Shiny Mathimals: _____

My strongest Mathimal:

_____

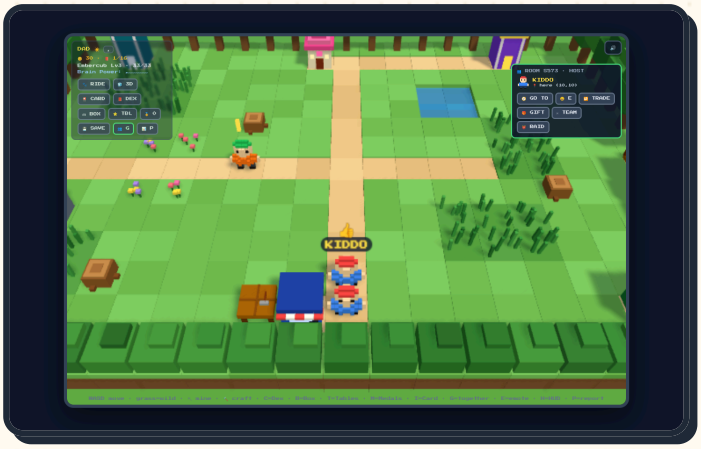
📝 Save code notes

Write down the date when you make a save code (💾 SAVE → ⬇ DOWNLOAD keeps a copy on the computer too).

👥 Playing Together

🔗 Linking up (1 minute)

1. Open the game on **both laptops** (internet needed).
2. Laptop 1: press G → **HOST A GAME**. A **4-letter code** appears.
3. Laptop 2: press G, type the code, press **JOIN**.
4. Your partner's panel appears top-right. You're linked!



Your partner walks around in your world, with their name above them.

🧭 What you can do

- ✔ 🧭 **GO TO** jumps right next to your partner.
- ✔ 😄 **Emotes:** E then 1–6 👋 😄 👍 ❤️ 🎉 ❓
- ✔ 🎁 **GIFT** sends coins, potions or balls from your own bag.
- ✔ 🔁 **TRADE** swaps one Mathimal each. ✨ One that hasn't evolved yet **evolves the moment it arrives!**

Battling together



Both answered within 3 seconds: TEAM-UP ATTACK!

⚔ Team battle

One tough wild Mathimal, **one shared HP bar**. You each answer **your own** questions.

👹 Raid boss

A **giant** boss and a **3-minute clock**. Win and it **joins both your teams**!

🤝 TEAM-UP attack

Both right within **3 seconds** = a bonus hit worth half of both attacks.

For Grown-ups

💾 Saving

- ✔ **Auto-save** happens in the background, in this browser on this computer. Open the game and press **CONTINUE ADVENTURE**.
- ✔ **New Adventure** asks first, and the replaced game can be brought back with **Restore Previous Adventure**.
- ✔ **Save codes** (💾 SAVE) hold the whole adventure. Copy or ⬇ download one now and then — they're the only backup if the browser data is cleared, and they move a game to another computer.
- ✔ A **red warning** means the browser isn't saving (e.g. private window). Make a save code straight away.

🖥️ Running the game

- ✔ It's one file: **mathimal-quest.html**. Double-click to open in Chrome or Edge. Works offline.
- ✔ Designed for a laptop or desktop with a keyboard.
- ✔ If it feels slow, press **3D** on the HUD to switch to the simpler 2D view.
- ✔ Updating: replace the file with the new version in the same place — progress carries over.

🧮 How the maths adapts

Maths Power (1–10) sets the difficulty: from simple addition up to times tables, division, missing numbers and word problems. It only rises on answers that are correct *and* reasonably quick (under 12 s, 18 s for word problems), so it tracks fluency rather than guessing. Two misses in a row ease it down.

Topic gyms (money, time, fractions, place value) each have their own 3 levels that step up with practice, so a new topic always starts gently.

The training report (P) shows accuracy and speed by question type.

Grown-up mode: choose 🧠 **General Knowledge** at the start (or on the Trainer Card) and battles ask quiz questions instead: 262 of them in three tiers, about NZ, te reo Māori, science, history, sport, the arts and more. They get harder as Brain Power rises.

🎁 Reward codes

Parent reward codes give coins or items when typed at the shop counter. Each works once. Keep them for real-life wins — homework done, a kind deed, reading time.

ℹ️ About

Mathimal Quest v5.0. All creatures and characters are original. PokeJosh was designed by the young trainer this game was made for.